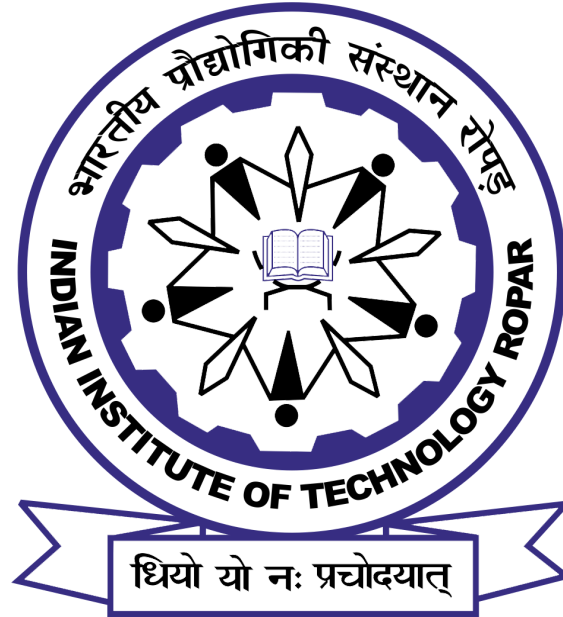


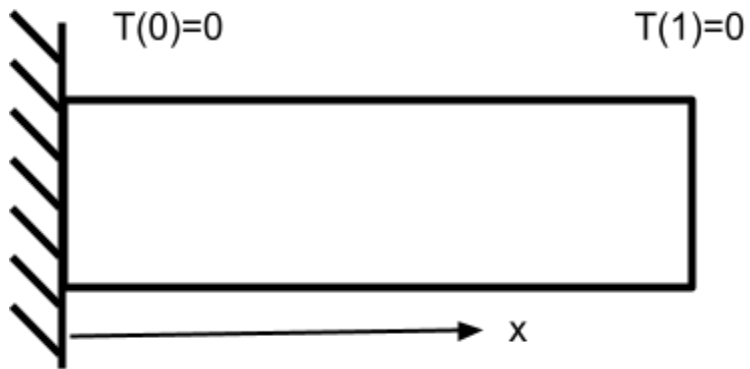
ME-504: Deep Learning For Physical Systems

Assignment 2



Submitted by-
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2025MEM1003

Q. Test Case-2



Solution -

The values of $k=3$, $q_0=20$ and total length of rod to be of 1 unit.

The number of values have been taken as “30” with 80% being used for training and the rest relegated to test the resultant.

Below page shows derivation of the temperature at different points (also taken as random).

Polynomial regression

Two values $\rightarrow$ actual $\bar{y}$
predicted $\hat{y}$

actual $\rightarrow k \frac{d^2 T}{dx^2} + q = 0$ for $0 \leq x \leq 1$
 given $q = q_0 \sin \frac{\pi x}{L}$ let $L = 1 \text{ m}$

$$k \frac{d^2 T}{dx^2} = -q_0 \sin \frac{\pi x}{L}$$

int $\rightarrow k \frac{dT}{dx} = -q_0 \cos \frac{\pi x}{L} \left(\frac{\pi}{L} \right) + C_1$

Given

$$T(0) = 0$$

$$T(1) = 0$$

int $\rightarrow k T = \frac{q_0}{L^2} \sin \frac{\pi x}{L} + C_1 x + C_2$

$$T = \frac{q_0 L^2}{\pi^2 k} \sin \frac{\pi x}{L} + \frac{C_1 x}{k} + \frac{C_2}{k}$$

$$T(0) = C_2 = 0$$

$$T(1) = C_1 = 0 \quad (\sin \pi = 0)$$

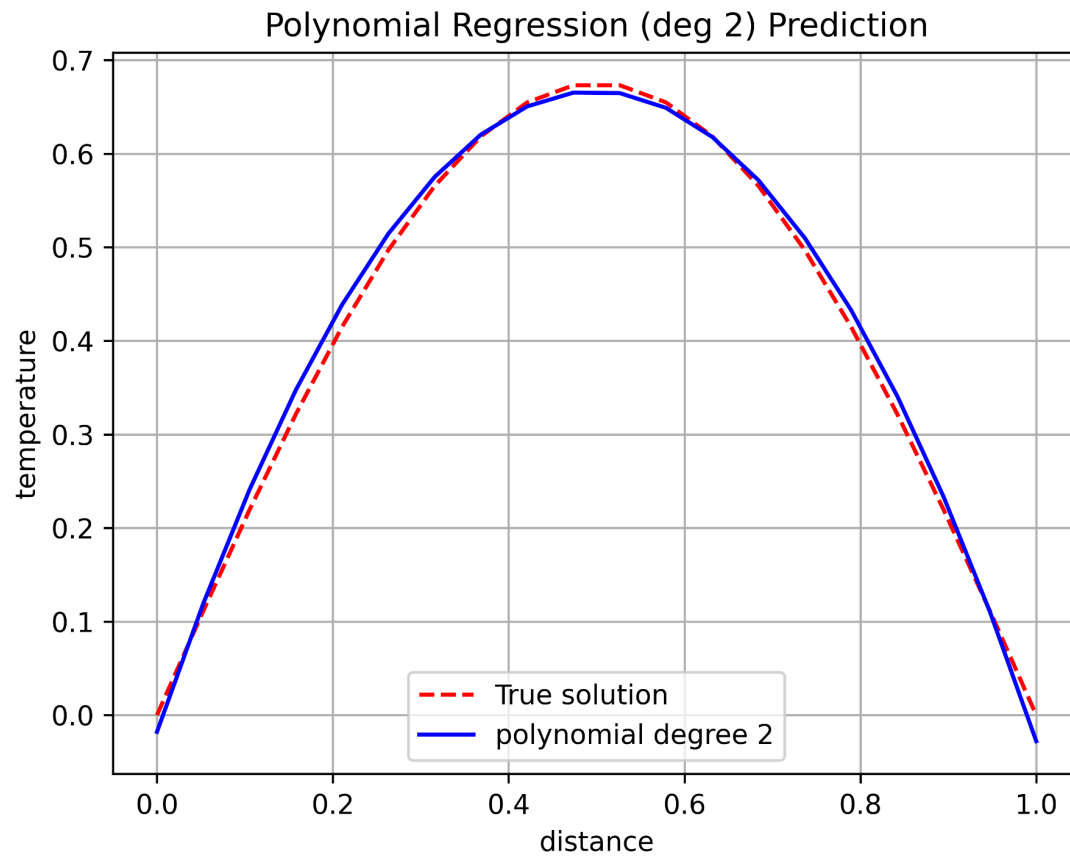
So $\bar{y} = \frac{q_0 L^2}{\pi^2 k} \sin \left(\frac{\pi x}{L} \right)$

predicted $\rightarrow \hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3 \dots w_n x^n$

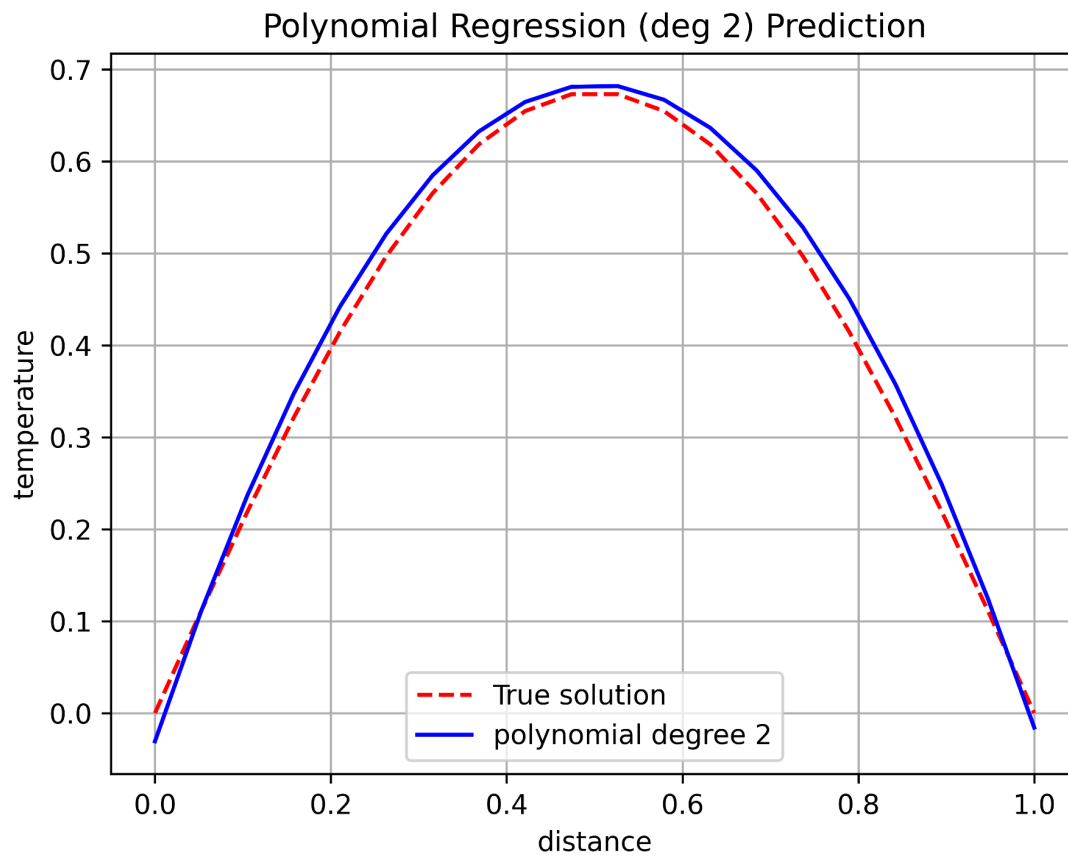
$$\hat{y} = \begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \dots \\ w_n \end{bmatrix}$$

$\underbrace{\hspace{10em}}_X \quad \underbrace{\hspace{2em}}_w$

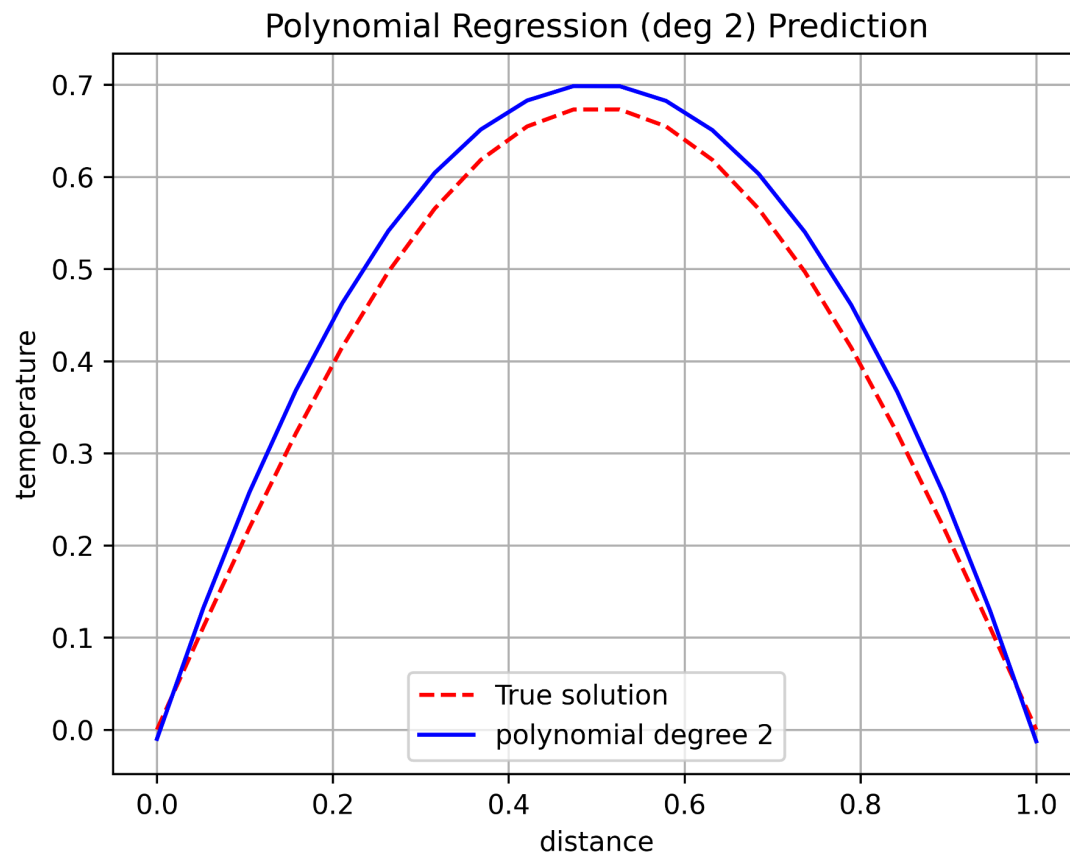
Output-



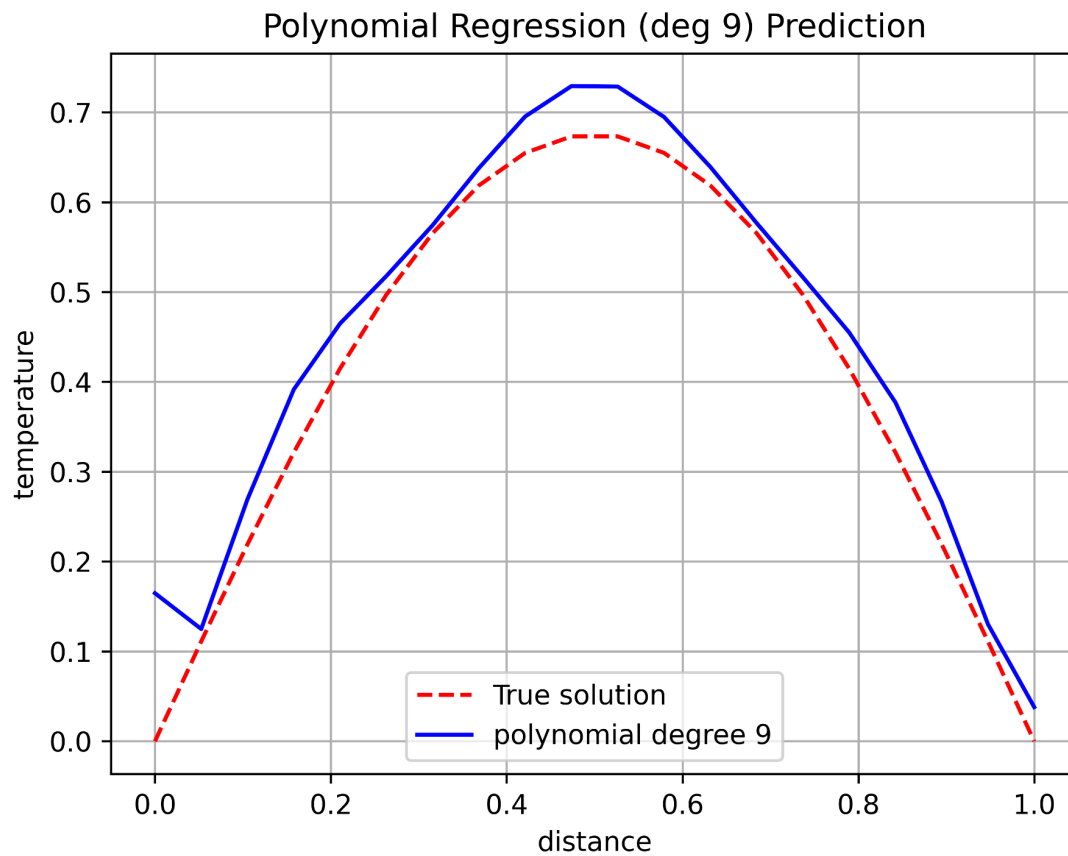
Noise set at 2% with 2nd degree polynomial. The polynomial approximation fits closely to the true solution



Noise is at 5% with the same polynomial fitted. Not much difference noted, though the fit is overextended.



Noise is at 10%, the polynomial model doesn't fit the true solution and exhibits overfitting.



Noise at 2% with degree of polynomial taken as 9, as can be observed, the solution doesn't fit properly, overfitting starts immediately not even satisfying the boundary conditions.